

Jun14Problems

le06 · 14.06.26



```
In [2]:
def remove_even_indices(l):
    return [l[i] for i in range(len(l)) if i % 2 == 1]

print(remove_even_indices([1,2,3,4,5,6]))
```

```
Out [2]:
[2, 4, 6]
```

```
In [3]:
def median_list(l):
    l = sorted(l)
    n = len(l)
    if n % 2:
        return l[n//2]
    return (l[n//2-1] + l[n//2]) / 2

print(median_list([1,2,3,4,5]))
print(median_list([1,2,3,4,5,6]))
```

```
Out [3]:
3
3.5
```

```
In [4]:
def longest_consecutive_sequence(l):
    s = set(l)
    best = []
    for x in s:
        if x - 1 not in s:
            cur = []
            y = x
            while y in s:
                cur.append(y)
                y += 1
            if len(cur) > len(best):
                best = cur
    print(len(best))
    print(*best)

longest_consecutive_sequence([100,4,200,1,3,2])
longest_consecutive_sequence([20,19,18,17,16])
longest_consecutive_sequence([10,3,7,2,1,8,6,4,11,5])
longest_consecutive_sequence([10,1,2,11,15])
```

```
Out [4]:
4
1 2 3 4
5
16 17 18 19 20
8
1 2 3 4 5 6 7 8
2
1 2
```

```
In [5]:
def remove_empty_strings(lst):
    return [x for x in lst if x]

print(remove_empty_strings(["Mike", "", "Emma", "Kelly", "", "Brad"]))
```

```
Out [5]:
['Mike', 'Emma', 'Kelly', 'Brad']
```

```
In [6]:
def common_elements(a, b):
    return list(set(a) & set(b))

print(sorted(common_elements([1,2,3,4,5], [3,4,5,6,7])))
```

```
Out [6]:
[3, 4, 5]
```

```
In [7]:
def rotate_right(lst, n):
    if not lst:
        return lst
    n %= len(lst)
    return lst[-n:] + lst[:-n]

print(rotate_right([1,2,3,4,5], 2))
print(rotate_right([10,20,30], 1))
print(rotate_right([1,2,3], 3))
print(rotate_right([7], 10))
print(rotate_right([1,2,3,4], 0))
print(rotate_right([1,2,3,4], 6))
```

```
Out [7]:
[4, 5, 1, 2, 3]
[30, 10, 20]
[1, 2, 3]
[7]
[1, 2, 3, 4]
[3, 4, 1, 2]
```